

Unified Wave-Pyramid Economic Models

Incorporating Supply-Chain Depth Dynamics

Abstract

This document combines the Unified Wave-Pyramid Economic Models framework with its extension to supply-chain depth. The combined framework traces value from raw material extraction through processing stages, incorporating binomial scarcity, inverse-exponential abundance decay, plus-sum labour dynamics, and zero-sum extraction pyramids. The unified master equation governs all models, with labour as the distinguishing factor between plus-sum and zero-sum systems. The supply-chain depth model parameterizes value accumulation across processing stages, showing how scarcity compounds through downstream amplification while labour adds plus-sum value at each stage.

Contents

1 Foundations: Notation and Shared Primitives

All models share a common set of primitives.

Definition 1 (Wave index). *A wave $\mathcal{W}_k$, $k = 1, 2, \dots, K$, is a decade-band of $n = 10$ binomial trials. The wave index acts as a slow-time variable layered above the fast trial index n .*

Definition 2 (Binomial access trial). *In each trial n , resource extraction either succeeds ($X_n = 1$) or fails ($X_n = 0$) with failure probability $p \in (0, 1)$. Trials are i.i.d. Bernoulli($1 - p$). The cumulative success count after n trials is*

$$S_n = \sum_{i=1}^n X_i \sim \text{Binomial}(n, 1 - p).$$

Definition 3 (Adjusted failure probability across waves). *Infrastructure maturation reduces p geometrically across waves:*

$$p_k = p_0 \cdot \theta^{k-1}, \quad \theta \in (0, 1).$$

Definition 4 (Access threshold n^*). *The access threshold n^* is the trial at which the commodity transitions from scarcity to abundance. It satisfies:*

$$\mathbb{E}[S_{n^*}] = n^*(1 - p) \geq D_{\min}.$$

2 Master Equation: Unified Synthesis

The four models are unified by the observation that every stock—commodity price, population, GDP, and speculative pool—obeys a logistic evolution modulated by a binomial access function and a fatigue (or extraction) drag:

$$X(t+1) = X(t) + \underbrace{B(n, p(t)) \cdot \Phi_{\text{access}}(t)}_{\text{binomial uplift}} \cdot \underbrace{\left(1 - \frac{X(t)}{K(t)}\right)}_{\text{logistic ceiling}} - \underbrace{E(t)}_{\text{extraction / fatigue drag}}, \quad (1)$$

where:

- $B(n, p(t))$ is the binomial uplift (access trials succeeding);
- $\Phi_{\text{access}}(t) = e^{-\lambda \max(0, t-t^*)}$ is the post-peak inverse-exponential dispersal;
- $K(t)$ is the wave-dependent carrying capacity (infrastructure $\times$ resource density);
- $E(t)$ is the extraction term: house edge in Model IV, resource depletion in Models I–II, automation drag in Model III.

3 Model I: Binomial Commodity Pricing

3.1 Scarcity Regime ($n < n^*$)

When extraction is difficult, each failed trial raises the marginal cost:

$$C(n) = C_0 \cdot (1 - p_k)^{-n},$$

with expected cost:

$$\mathbb{E}[C(n)] = C_0 \cdot \frac{1}{(1 - p_k)^n}.$$

3.2 Abundance Regime: Inverse-Exponential Decay ($n \geq n^*$)

Once the access threshold is crossed:

$$P(t) = P_{\text{peak}} \cdot e^{-\lambda t} + P_{\text{floor}}, \quad t = n - n^* \geq 0,$$

where

$$P_{\text{floor}} = \frac{\alpha_0}{I \cdot \rho}$$

is the infrastructure-and-density-constrained price floor. Here $I \in (0, 1]$ is infrastructure quality and $\rho \in (0, 1]$ is raw material density.

3.3 Consumer Fatigue and Plateau Phase

Logistic damping enters once adoption saturates. Define fatigue onset as $n_f = f \cdot n_{\text{max}}$ for $f \in (0, 1)$. For $n > n_f$:

$$P(n) = P_{\text{floor}} + [P_{\text{peak}} e^{-\lambda(n-n^*)}] \cdot (1 + \mu\phi^2), \quad \phi = \frac{n - n_f}{n_{\text{max}} - n_f}.$$

Demand follows:

$$D(n) = \begin{cases} D_{\text{max}}(1 - e^{-\delta(n-n^*)}) & n \in [n^*, n_f], \\ D_{\text{max}}(1 - \frac{1}{2}\phi^2) & n > n_f. \end{cases}$$

4 Model II: Population Dynamics as Commodity Access

4.1 Core Isomorphism

Population dynamics are governed by access to carrying-capacity commodities: calories, medicine, shelter, and energy. The commodity is carrying capacity $K(t)$.

4.2 Scarcity Era (Malthusian Regime)

For $w < w^*$:

$$K(w) = K_{\text{sc}}(1 + \epsilon w), \quad \epsilon \ll 1,$$

and population evolves under a logistic ceiling:

$$N(w + 1) = N(w)(1 + r_w) \min \left(1, \frac{K(w)}{N(w)} \right).$$

4.3 Access Boom (Inverse-Exponential Carrying Capacity)

Post access-wave:

$$K(w) = K_{\text{sc}} + (K_{\infty} - K_{\text{sc}}) (1 - e^{-\lambda I \rho (w - w^*)}),$$

with population growth:

$$N(w + 1) = N(w) + N(w)g(w) \left(1 - \frac{N(w)}{K(w)}\right),$$

where

$$g(w) = \lambda I \rho \cdot \left(1 - \frac{N(w)}{K(w)}\right) \cdot 1.5.$$

4.4 Demographic Transition (Fatigue Analog)

For $w > w_f$:

$$g(w) = \lambda \cdot 0.3 \cdot (1 - \phi) \cdot I \rho - \delta_d \phi, \quad \phi = \frac{w - w_f}{K - w_f}.$$

4.5 Infrastructure Dependence of the Population Ceiling

The maximum achievable population:

$$N_{\text{max}} = K_{\infty}(I, \rho) = K_{\text{ref}} \cdot I^{\alpha} \cdot \rho^{\beta}.$$

5 Model III: Plus-Sum Growth, Labour, and the Myth of Endless Growth

5.1 Four Coupled Stocks

Let t index decades from 1800. Define:

$R(t)$ = resource access index $\in [0, 1]$, $L(t)$ = labour pool (normalised), $G(t)$ = GDP index, $K(t)$ =

5.2 Resource Dynamics

$$R(t) = \begin{cases} R_0 \cdot (1 - p_{[t/10]})^{-t} \cdot c & t < t^*, \\ R_{\text{peak}} e^{-\lambda_R (t - t^*)} & t \geq t^*. \end{cases}$$

5.3 Labour Pool with GDP Feedback

$$L(t + 1) = L(t) \left(1 + r_L + \gamma \frac{G(t)}{K_{\infty}^G} \cdot e^{-\delta_L \max(0, t - t_f)}\right).$$

5.4 GDP: The Plus-Sum Identity with Resource Drag

$$G(t+1) = G(t) + \Delta G_R(t) + \Delta G_L(t) + \Delta G_K(t) + \Delta G_A(t) \cdot \left(1 - \frac{G(t)}{K_G^G}\right),$$

with components:

$$\begin{aligned}\Delta G_R(t) &= G(t) \cdot R(t) \cdot \eta_R, \\ \Delta G_L(t) &= G(t) \cdot \frac{L(t+1) - L(t)}{L(t)} \cdot \alpha \left(1 - \beta \frac{t}{T}\right), \\ \Delta G_K(t) &= K(t) \cdot \rho_K, \\ \Delta G_A(t) &= -G(t) \cdot \delta_A \cdot \min\left(1, \frac{t}{T} \cdot 1.5\right).\end{aligned}$$

5.5 The Myth Gap

Naive extrapolation:

$$G_{\text{naive}}(t) = G(t_0) \cdot (1 + \bar{g})^{t-t_0}.$$

The myth gap:

$$\mathcal{M}(t) = G_{\text{naive}}(t) - G(t) \rightarrow K_\infty^G - G(\infty) \quad \text{as } t \rightarrow \infty.$$

5.6 Capital Accumulation

$$K(t+1) = K(t)(1 + \rho_K/2) + G(t) \cdot s_K.$$

6 Model IV: Casino Economics and Financial Bubble Pyramid

6.1 Zero-Sum Identity

Theorem 1 (Zero-sum conservation law). *In any closed wagering or speculative pool $\mathcal{P}$ of size W_0 with no external value creation:*

$$\sum_{i=1}^N \ell_i(t) \equiv \mathcal{H}(t) \quad \forall t,$$

where ℓ_i is the loss of player i and $\mathcal{H}(t)$ is cumulative house extraction.

6.2 Pyramid Layer Structure

The casino and financial bubble share identical extraction architecture:

- Retail Base (Slot Layer): $E_{\text{ret}}(n) = W_{\text{ret}} \cdot h_{\text{ret}}$
- Table/Mid-Tier Layer with Leverage: $E_{\text{mid}} = W_{\text{mid}} \cdot h_{\text{mid}} + \Lambda \cdot r_{\text{spread}}$
- Debt Superstructure: $E_{\text{debt}}(t) = \sum_{j=1}^{N_d} L_j \cdot r_j(t)$
- House Short Position: $\Pi_{\text{short}} = \sigma \cdot W_0 \cdot \frac{P^* - P_{\text{crash}}}{P^*}$

6.3 Pool Depletion Dynamics

$$\mathcal{P}(t+1) = \mathcal{P}(t)(1 + \psi_t) - E(t),$$

where hype inflow rate:

$$\psi_t = \begin{cases} \psi_0 \cdot 1.2 & \text{accumulation,} \\ \psi_0 \cdot 1.0 & \text{euphoria,} \\ \psi_0 \cdot 0.6 & \text{distribution,} \\ 0 & \text{collapse.} \end{cases}$$

Collapse triggered when:

$$\frac{\Lambda(t)}{\mathcal{P}(t)} \geq \tau \Rightarrow \text{cascade liquidations.}$$

7 Supply-Chain Depth Model

7.1 Motivation and Scope

Real goods reaching consumers are not single-stage commodities—they are the output of a chain of transformations. We formalise this as the supply-chain depth model, parameterised by $d \in \mathbb{Z}_{>0}$, the number of distinct processing stages.

Product class	d	Raw material inputs	Chains
Fresh produce	1-2	Soil, water, seed	1
Processed agriculture	2-4	Grain, energy, water	2-3
Construction material	4-7	Limestone, clay, fuel	3-5
Complex manufactured	8-15	Ore, silicon, hydrocarbons	6-12
Advanced technology	12-25+	Rare earths, specialty gas	10-20

Table 1: Canonical product classes

7.2 Multi-Stage Supply-Chain Model

7.2.1 Stage Definition and Notation

Each stage $k \in \{1, \dots, d\}$ is characterised by:

- p_k : binomial failure probability
- ℓ_k : labour hours per unit output
- w_k : wage rate (plus-sum: includes skill premium)
- d_k : logistic distance between stage $k-1$ and stage k
- τ_k : transport cost per unit per km
- m_k : material conversion ratio (units input per unit output)
- η_k : energy intensity

- e_k : energy price
- μ_k : stage markup

7.2.2 Stage-by-Stage Cost Propagation

Let C_k denote the unit cost at the output of stage k , with $C_0 \equiv C_0(n, p)$ as raw material extraction cost.

$$C_k = m_k C_{k-1} \cdot (1 - p_k)^{-n_k} + \underbrace{\ell_k w_k}_{\text{labour (plus-sum)}} + \underbrace{\eta_k e_k}_{\text{energy}} + \underbrace{\tau_k d_k}_{\text{logistics}} + \underbrace{\mu_k C_{k-1}}_{\text{stage margin}}.$$

Final consumer price:

$$P_{\text{fin}} = C_d \cdot (1 + \mu_{\text{retail}}).$$

7.2.3 Recursive Expansion

$$P_{\text{fin}} = (1 + \mu_{\text{ret}}) \left[\underbrace{\prod_{k=1}^d (m_k (1 - p_k)^{-n_k} + \mu_k)}_{\text{scarcity-margin compounding}} \cdot C_0 + \sum_{k=1}^d \underbrace{\prod_{j=k+1}^d (m_j (1 - p_j)^{-n_j} + \mu_j)}_{\text{downstream amplification of stage-k cost}} \cdot (\ell_k w_k + \eta_k e_k + \tau_k d_k) \right].$$

7.3 Plus-Sum Labour Across the Chain

Total plus-sum value added by labour:

$$\mathcal{V}_L = \sum_{k=1}^d \ell_k w_k \cdot \prod_{j=k+1}^d (m_j (1 - p_j)^{-n_j} + \mu_j).$$

Raw material pass-through:

$$\phi_{\text{raw}} = \frac{\prod_{k=1}^d (m_k (1 - p_k)^{-n_k} + \mu_k) \cdot C_0}{P_{\text{fin}} / (1 + \mu_{\text{ret}})}.$$

7.4 Logistic Distance Penalty

$$L_{\text{fin}} = \sum_{k=1}^d \tau_k d_k \cdot \prod_{j=k+1}^d (m_j (1 - p_j)^{-n_j} + \mu_j).$$

7.5 Canonical Product Classes

7.5.1 Class I — Fresh Produce ($d = 1 - 2$)

$$C_1 = m_1 C_0 (1 - p_1)^{-n_1} + \ell_1 w_1 + \eta_1 e_1,$$

$$C_2 = m_2 C_1 (1 - p_2)^{-n_2} + \tau_2 d_2,$$

$$P_{\text{fin}} = C_2 (1 + \mu_{\text{ret}}).$$

7.5.2 Class II — Concrete ($d = 4 - 7$)

Five primary stages: limestone quarrying, crushing & grinding, kiln clinker, cement milling, batching plant, pour & cure.

Key features:

- Energy intensity at kiln stage dominates: $\eta_2 e_2 \gg \ell_2 w_2$
- Multiple parallel raw material chains
- Low labour share: $\mathcal{V}_L / P_{\text{fin}} \approx 15 - 25\%$

7.5.3 Class III — Complex Manufactured Goods ($d = 8 - 25+$)

DAG (directed acyclic graph) of parallel sub-chains converging at assembly:

$$C_{\text{asm}} = \sum_{i \in \mathcal{T}} q_i C_{d_i}^{(i)} + \ell_{\text{asm}} w_{\text{asm}} + \tau_{\text{asm}} d_{\text{asm}}.$$

7.6 Labour Share as a Function of Depth

$$\Lambda(d) = \frac{\mathcal{V}_L}{P_{\text{fin}}/(1 + \mu_{\text{ret}})} = \frac{\sum_{k=1}^d \ell_k w_k \prod_{j=k+1}^d (m_j(1 - p_j)^{-n_j} + \mu_j)}{C_d}.$$

Theorem 2 (Labour share monotonicity). *If all stages have $p_k < 1/2$, $m_k \leq 2$, and w_k is non-decreasing in k , then $\Lambda(d)$ is non-decreasing in d for fixed C_0 .*

7.7 Disruption Propagation

Single-stage disruption:

$$\frac{\partial P_{\text{fin}}}{\partial p_k} = m_k n_k (1 - p_k)^{-n_k - 1} C_{k-1} \cdot \prod_{j=k+1}^d (m_j(1 - p_j)^{-n_j} + \mu_j) \cdot (1 + \mu_{\text{ret}}).$$

Correlated disruption across sub-chains:

$$\text{Var} \left[\sum_i q_i \Delta C^{(i)} \right] = \sum_i q_i^2 \text{Var}[\Delta C^{(i)}] + 2 \sum_{i < j} q_i q_j \rho_{ij} \sqrt{\text{Var}[\Delta C^{(i)}] \text{Var}[\Delta C^{(j)}]}.$$

8 Cross-Model Identities

$$P_{\text{floor}} = \frac{\alpha_0}{I \cdot \rho} \equiv \frac{G_{\text{min}}}{N_{\text{max}}} \quad (\text{commodity floor} = \text{subsistence per capita}),$$

$$K_{\infty}^{\text{pop}} = K_{\text{ref}} \cdot I^{\alpha} \cdot \rho^{\beta} \equiv K_{\infty}^G / \alpha_{\text{GDP}} \quad (\text{population and GDP ceilings share same } I, \rho \text{ inputs}),$$

$$\mathcal{M}(t) = G_{\text{naive}}(t) - G(t) \equiv \frac{P(0) - P(t)}{P(0)} \cdot W_0 \quad (\text{myth gap} \sim \text{pool depletion in bubble}),$$

$$E_{\text{ret}} = W_0 \cdot s \cdot h \equiv N \cdot \mathbb{E}[\ell_i] \quad (\text{aggregate extraction} = \text{average individual loss}).$$

8.1 Labour as the Distinguishing Factor

The sole structural difference between plus-sum (Models I-III) and zero-sum (Model IV) systems is the sign of ΔG_L :

$$\begin{cases} \Delta G_L > 0 & \text{plus-sum: labour creates new value,} \\ \Delta G_L = 0 & \text{zero-sum: no production, only redistribution.} \end{cases}$$

9 Consumer Price Decomposition Identity

For any product of depth d , the final price satisfies:

$$P_{\text{fin}} = (1 + \mu_{\text{ret}}) \left[\underbrace{\Phi_{\text{raw}} \cdot C_0}_{\text{material origin}} + \underbrace{\mathcal{V}_L}_{\text{plus-sum labour}} + \underbrace{L_{\text{fin}}}_{\text{logistics}} + \underbrace{\mathcal{E}}_{\text{energy stack}} + \underbrace{M}_{\text{margin stack}} \right],$$

where $\Phi_{\text{raw}} = \prod_{k=1}^d (m_k(1 - p_k)^{-n_k} + \mu_k)$ is the scarcity-compounding factor.

10 Cross-Class Comparison

Property	Fresh produce	Concrete	Automobile	Smartphone
Depth d	1-2	4-7	8-12	12-25+
Sub-chains $ \mathcal{T} $	1	3-4	50-200	200-1,000
ϕ_{raw} (material share)	15-30%	25-40%	8-15%	3-8%
Λ (labour share)	40-60%	15-25%	30-45%	20-40%
Dominant cost term	$\ell_k w_k, \tau d$	ηe (kiln)	Assembly ℓw	R&D + IP
Logistic sensitivity	Very (perishable)	High (chemical)	Medium	Low (air freight)
Scarcity amplifier	Weather (Δp volatile)	Limestone (low p)	Steel/chips	Rare earths (low p)
Automation a_k effect	Moderate	High at $k = 2, 3$	High at assembly	Very high at assembly

Table 2: Analytical comparison across product classes

11 Parameter Reference Tables

11.1 Unified Model Parameters

11.2 Supply-Chain Depth Parameters

12 Concluding Remarks

The master equation makes explicit what each model shares: a binomial access process whose success probability declines with infrastructure and material scarcity; an inverse-exponential post-peak dispersal; a logistic ceiling set by processed infrastructure and resource density; and an extraction or fatigue drag.

Symbol	Model	Definition	Typical range
p_0	I-IV	Baseline binomial failure probability	[0.05, 0.95]
θ	I-IV	Wave-based p decay factor	[0.80, 0.95]
n^*	I-III	Access threshold (trials)	[5, 60]
λ	I-IV	Inverse-exp decay coefficient	[0.01, 0.40]
I	I-III	Infrastructure quality	(0, 1]
ρ	I-III	Raw material density	(0, 1]
P_{floor}	I	Commodity price floor	$\alpha_0/(I\rho)$
α	III	Labour plus-sum productivity	[0.5, 3.0]
β	III	Post-industrial labour drag	[0.1, 1.0]
γ	II-III	GDP-to-birth-rate elasticity	[0.0, 1.0]
K_∞^G	III	GDP carrying capacity	application-specific
δ_A	III	Automation displacement rate	[0.0, 0.15]
h	IV	House edge (casino/bubble)	[0.005, 0.35]
ξ	IV	Leverage ratio	[1, 20]
σ	IV	House short fraction of pool	[0, 0.5]
τ	IV	Collapse trigger (leverage ratio)	[2, 15]
ψ	IV	Hype inflow rate	[0.02, 0.40]
$\mathcal{M}(t)$	III-IV	Myth gap: naive minus actual	R

The myth of endless growth emerges not from irrationality but from structural extrapolation: when resource, labour, and capital inputs each genuinely accelerate during their access waves, the trend line is correct. The error is in treating the logistic ceiling K_∞^G as infinite. The Haber-Bosch moment, the Green Revolution, the demographic dividend, and the post-war capital boom each contributed a real, finite uplift to GDP. Their exhaustion—resource dispersal λ , demographic contraction δ_d , and automation drag δ_A —was always present in the equations; it was invisible only while the access-phase gain term dominated.

The financial bubble is the limiting case where the plus-sum labour term vanishes entirely: $\Delta G_L = 0$. The pyramid then extracts from its own base until the leverage-to-pool ratio exceeds τ and cascade liquidation collapses the structure—a process perfectly isomorphic to the binomial failure rate $p \rightarrow 1$ in the commodity scarcity regime, but operating in reverse: not the cost of extraction rising, but the cost of repayment consuming the remaining pool.

The supply-chain depth model makes two structural claims that generalise the original binomial commodity framework. First, scarcity compounds across stages. A disruption to an upstream raw material does not simply raise that material’s price in isolation—it is amplified by every downstream conversion ratio, margin, and scarcity factor between the point of disruption and the consumer. This compounding is superlinear in the depth d , explaining why rare earth shortages in the mining sector produce semiconductor shortages in finished electronics within months.

Second, labour is the countervailing plus-sum force. At every stage, $\ell_k w_k$ enters the cost stack not as a redistribution of existing value but as new value created by human time and skill applied to capital. Deep chains (d large) therefore contain more accumulated plus-sum value than shallow chains—a fact obscured by the visibility of raw material prices (which are quoted on commodity exchanges) versus labour’s diffuse contribution across many invisible processing stages.

Symbol	Domain	Definition	Typical range
d	$\mathbb{Z}_{>0}$	Supply-chain depth (stages)	1-25+
p_k	$(0, 1)$	Stage-k binomial failure probability	$[0.01, 0.6]$
n_k	$\mathbb{Z}_{>0}$	Trials per stage	1-50
m_k	$[1, \infty)$	Material conversion ratio (input/output)	1.0-3.0
ℓ_k	$\mathbb{R}_{>0}$	Labour hours per unit at stage k	0-100
w_k	$\mathbb{R}_{>0}$	Wage rate at stage k (\$/hr)	2-500
σ	$\mathbb{R}_{>0}$	Skill premium exponent	0-3
η_k	$\mathbb{R}_{>0}$	Energy intensity (kWh/unit)	0-10,000
e_k	$\mathbb{R}_{>0}$	Energy price at stage k (\$/kWh)	0.05-0.50
d_k	$\mathbb{R}_{>0}$	Logistic distance (km)	0-20,000
τ_k	$\mathbb{R}_{>0}$	Transport cost (\$/unit/km)	10^{-5} - 10^{-2}
μ_k	$\mathbb{R}_{>0}$	Stage markup (fraction of input cost)	0.02-0.40
μ_{ret}	$\mathbb{R}_{>0}$	Retail margin	0.10-0.60
a_k	$[0, 1]$	Automation penetration at stage k	0-0.95
β_{mach}	$\mathbb{R}_{>0}$	Machine energy overhead factor	0.1-1.0
ϕ_{raw}	$(0, 1)$	Raw material share of final price	0.03-0.50
$\Lambda(d)$	$(0, 1)$	Labour share of final price	0.10-0.70
Φ_{raw}	$[1, \infty)$	Scarcity-compounding factor	1 - 10^6
$\mathcal{V}_L$	$\mathbb{R}_{>0}$	Total plus-sum labour value added	—
L_{fin}	$\mathbb{R}_{>0}$	Total logistics cost (amplified)	—

The interplay between these two forces—scarcity compounding Φ_{raw} amplifying upstream shocks, and labour value $\mathcal{V}_L$ growing with chain depth—determines the resilience and price structure of every manufactured good. Products high in both (rare-earth electronics) are simultaneously the most valuable and the most vulnerable to upstream disruption. Products high in labour but low in scarcity (fresh produce under favourable conditions) are vulnerable instead through the logistic channel L_{fin} , where distance and perishability are the binding constraints. Products high in energy and capital but low in labour (concrete, bulk steel) depend on the stability of energy prices e_k , which themselves follow a commodity price curve with their own binomial access wave.

In each case, the framework reduces to the same master equation: binomial access governs what flows in, plus-sum labour governs what value is added, and logistic distance governs how much of the surplus is consumed in transit before the consumer takes delivery.